

- 10 (a) (i) e.m. radiation / photons is produced whenever a charged particle is acceleratedM1
wavelength depends on magnitude of acceleration A1
electrons have a distribution of accelerations A1
so continuous spectrum A0 [3]
- (ii) *either* when electron loses all its energy in one collision
or when energy of electron produces a single photon B1 [1]
- (b) (i) parallel beam (in matter) B1
 $I = I_0 \exp(-\mu x)$ M1
 I , I_0 , (μ) and x explained A1 [3]
- (ii) *either* low-energy photons absorbed (much) more readily
or low-energy photons (far) less penetrating B1
low-energy photons do not contribute to X-ray image B1
low energy photons could cause tissue damage B1 [3]

[Total: 10]

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11 (a) amplitude modulation (allow AM)